# Explainability Methods for Graph Convolutional Neural Networks

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CONTENTS

# Motivation

Convolutional Neural Networks (CNNs)
: recent success in CV is mainly due to emergence of deep convolutional neural networks
e.g., object recognition, object detection, and semantic segmentation

The end-to-end learning strategy
: turned CNN into powerful data-driven tools for learning from a large corpus of visual data
: hinders the explainability and interpretability of decisions made by CNNs

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Increasing number of works studying the inner working + explaining the decisions of CNNs

- Contrastive Gradient-based Saliency Maps (CG)
- Class Activation Mapping (CAM)
- Excitation Backpropagation (EB)

# Motivation

Graph Convolutional Networks (GCNNs)

: successfully applied to many graph data on various learning tasks

e.g., node classification, graph classification, link prediction

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With the growing use of GCNNs comes the need for explainability

CNNs : designed for grid structured data (e.g., images in Euclidean spaces)

GCNNs : extension of CNNs to non-Euclidean space

(e.g., graphs - scene graph analysis, 3D-shape analysis, social networks, chemistry)

**" Adaptation of explainability methods for CNNs (CG, CAM, EB) to GCNNs "**

# Interpretability

**Interpretability** : about the extent to which a cause and effect can be observed within a system

**Explainability** : extent to which the internal mechanics of a machine or deep learning system can be explained in human terms

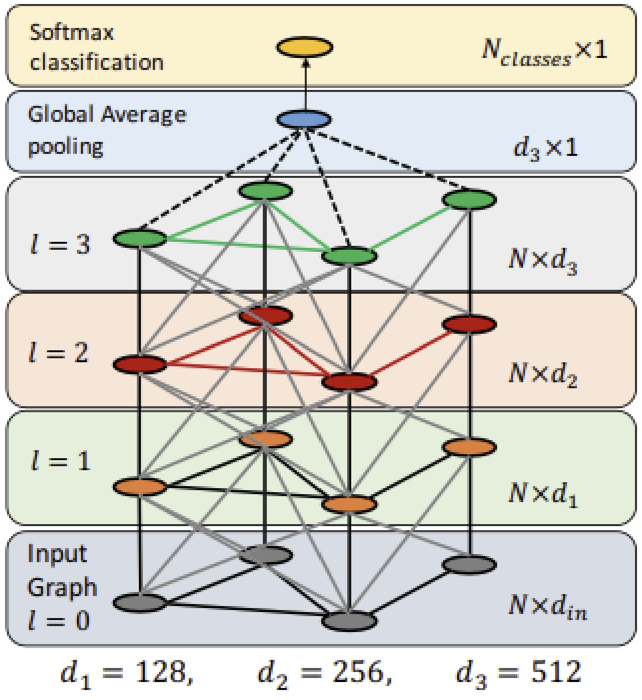
**Explainability** Methods for CNNs

: identify the important substructures of the input data for decision regarding a task

used as an explanatory tool or as a tool to discover unknown underlying substructures in the data

- Contrastive Gradient-based Saliency Maps (CG)
- Class Activation Mapping (CAM)
- Gradient-weighted Class Activation Mapping (Grad-CAM)
- Excitation Backpropagation (EB)

# Graph Convolutional Neural Networks




Inspired CNN successfully used in CV

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extend convolutional operations onto graph as an aggregation method

How to determine the **information** in the graph
: The value contained in each node (information contained in each node feature matrix) determines node information without changing the shape of the graph → Change the values in the node feature matrix to be updated

Recently found use in diverse **applications**
: super-pixel classification, classifying research papers from their citation network, N-grams for text categorization, shape segmentation, skeleton-based action recognition
scene graph classification, molecule classification

# Q & A

# Explainability for CNNs
## Contrasive Gradient-based

most straight-forward and well-established approach

simply differentiates the output of the model with respect to the model input → create heat-map
relevance importance = norm of the gradient over input variables

resulting gradient in the input space points in the direction corresponding to the maximum positive rate of change in the model output
(negative values in the gradient are discarded)

$$L^c_{Gradient} = \|\mathrm{ReLU}\left(\frac{\partial y^c}{\partial x}\right)\| \quad , \qquad (1)$$

# Explainability for CNNs
## Class Activation Mapping

### CAM

identify important, class-specific features at the last convolutional layer as opposed to the input space
such features tend to be more semantically meaningful (e.g., faces instead of edges)

requires the layer immediately before the softmax classifier (output layer) to be a convolutional layer followed by a global average pooling (GAP) layer

$$e_k = \frac{1}{Z}\sum_i \sum_j F_{k,i,j} \quad (2)$$

$$y^c = \sum_k w_k^c e_k, \quad (3)$$

$$L_{CAM}^c[i,j] = \text{ReLU}\left(\sum_k w_k^c F_{k,i,j}\right). \quad (4)$$

### Grad-CAM

improves upon CAM by relaxing the architectural restriction that the penultimate layer must be a convolution

use feature map weights that are based on back-propagated gradients

ReLU function ensures that only features that have a positive influence on the class prediction are non-zero

$$\alpha_k^c = \frac{1}{Z}\sum_i \sum_j \frac{\partial y^c}{\partial F_{k,i,j}}. \quad (5)$$

$$L_{Grad-CAM}^c[i,j] = \text{ReLU}\left(\sum_k \alpha_k^c F_{k,i,j}\right), \quad (6)$$

# Explainability for CNNs

## Excitation Backpropagation

simple, but empirically effective explanation method

demonstrated experimentally that explainability approaches such as EB, which ignore nonlinearities in the backward-pass through the network, are able to generate heat-maps that "conserve" evidence for or against a network predicting any particular class

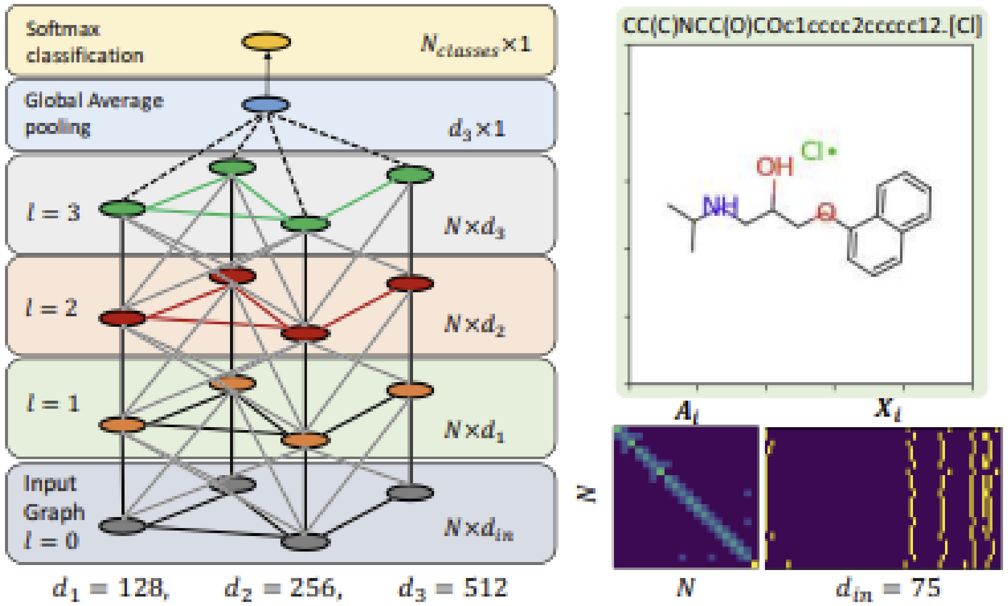
$$P(a_j^{(l-1)}) = \sum_i P(a_j^{(l-1)}|a_i^l)P(a_i^l). \qquad (7)$$

$$P(a_j^{(l-1)}|a_i^l) = \begin{cases} Z_i^{(l-1)} y_j^{(l-1)} W_{ji}^{(l-1)} & \text{if } W_{ji}^{(l-1)} \geq 0, \\ 0 & \text{otherwise,} \end{cases} \qquad (8)$$

where $Z_i^{(l-1)} = \left( \sum_j y_j^{(l-1)} W_{ji}^{(l-1)} \right)^{-1}$ is a normalization factor

EB generates a heat-map in the pixel-space w.r.t class c by staring with P = 1 at the output layer and applying Equation (7) recursively

# Graph Convolutional Neural Networks




GCNN + GAP architecture with the visualization of the input feature and adjacency matrix for sample molecule from the BBBP dataset

graph with N nodes be defined with its node attributes X and its adjacency matrix A

$$F^l(X, A) = \sigma(\underbrace{\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}}_{V} F^{(l-1)}(X, A) W^l), \qquad (9)$$

**Molecule Classification**

each molecule can be represented as an attributed graph

node feature : summarize the local chemical environment of atoms in the molecule (including atom-types, hybridization types, valence structures)

adjacency matrix : atomic bonds and demonstrate the connectivity of the whole molecule

label : indicating a certain chemical property (e.g., blood-brain-barrier penetrability or toxicity)

# Explainability for GCNNs
## Gradient-based

extension of CNN explainability methods to GCNNs

k'th graph convolutional feature map at layer l : $$F_k^l(X,A) = \sigma(VF^{(l-1)}(X,A)W_k^l) \quad (10)$$

GAP feature after the final convolutional layer L : $$e_k = \frac{1}{N}\sum_{n=1}^{N} F_{k,n}^L(X,A) \quad , \quad (11)$$

class score : $y^c = \sum_k w_k^c e_k.$

**Gradient based** heat-maps over nodes n : $$L_{Gradient}^c[n] = \|\mathrm{ReLU}\left(\frac{\partial y^c}{\partial X_n}\right)\| \quad . \quad (12)$$

**CAM** heat-maps : $$L_{CAM}^c[n] = \mathrm{ReLU}(\sum_k w_k^c F_{k,n}^L(X,A))) \, . \quad (13)$$

**Grad-CAM**'s class specific weights for class c at layer l, and for feature k : $$\alpha_k^{l,c} = \frac{1}{N}\sum_{n=1}^{N} \frac{\partial y^c}{\partial F_{k,n}^l} \quad , \quad (14)$$

heap-map calculated from layer l : $$L_{Grad-CAM}^c[l,n] = \mathrm{ReLU}(\sum_k \alpha_k^{l,c} F_{k,n}^l(X,A)) \, . \quad (15)$$

# Explainability for GCNNs
## Excitation Backpropagation

### Excitation Backpropagation's heat-map

equations for backward passes through the softmax classifier and the GAP layer

$$\begin{cases} p(e_k) = \sum_c \frac{e_k ReLU(w_k^c)}{\sum_k e_k ReLU(w_k^c)} p(c) & \text{Softmax} \\ p(F_{k,n}^L) = \frac{F_{k,n}^L}{Ne_k} p(e_k) & \text{GAP}, \end{cases} \quad (17)$$

for notional simplicity, we decompose a graph convolutional operator into

$$\begin{cases} \hat{F}_{k,n}^l = \sum_m V_{n,m} F_{k,m}^l \\ F_{k',n}^{(l+1)} = \sigma(\sum_{k'} \hat{F}_{k,n}^l W_{k,k'}^l), \end{cases} \quad (18)$$

corresponding backward passes for these two functions :

$$\begin{cases} p(F_{k,n}^l) = \sum_m \frac{V_{n,m} F_{k,n}^l}{\sum_n V_{n,m} F_{k,m}^l} p(\hat{F}_{k,m}^l) \\ p(\hat{F}_{k,n}^l) = \sum_{k'} \frac{\hat{F}_{k,n}^l ReLU(W_{k,k'}^l)}{\sum_k \hat{F}_{k,n}^l ReLU(W_{k,k'}^l)} p(F_{k',n}^{(l+1)}). \end{cases} \quad (19)$$

generate the heat-map over the input layer by recursively backpropagating through the network and averaging the backpropagated probability heat-maps on the input :

$$L_{EB}^c[n] = \frac{1}{d_{in}} \sum_{k=1}^{d_{in}} p(F_{k,n}^0) \quad . \quad (20)$$

# Q & A

# Explanations on Scene Graphs

Dataset : Visual Genome dataset

Construct two binary classification task on scene graphs from the Visual Genome

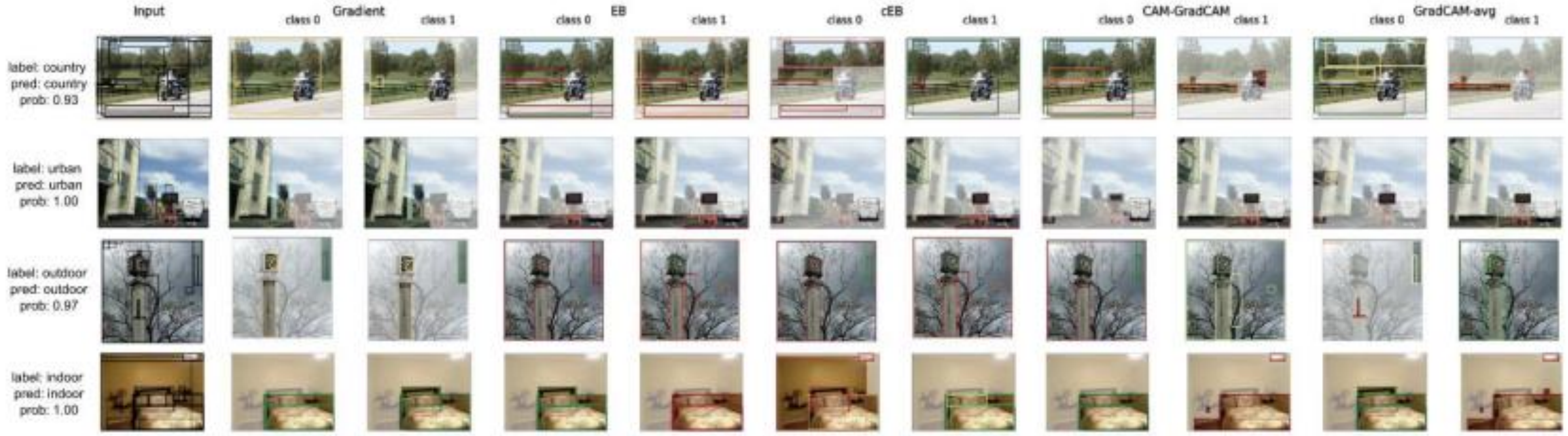
- "country" : country, countryside, farm, rural, cow, crops, sheep
- "urban" : urban, city, downtown

- "indoor" : indoor, room, office, bedroom, bathroom
- "outdoor" : outdoor, nature, outside

| Dataset | $N_p$ | $N_n$ | AUC-ROC | | AUC-PR | |
|---|---|---|---|---|---|---|
| | | | Train | Test | Train | Test |
| Country vs. Urban | 3267 | 3862 | 0.987 | 0.986 | 0.939 | 0.936 |
| Indoor vs. Outdoor | 8290 | 7971 | 0.962 | 0.963 | 0.871 | 0.876 |
| BBBP | 1560 | 470 | 0.991 | 0.966 | 0.994 | 0.966 |
| BACE | 691 | 821 | 0.994 | 0.966 | 0.939 | 0.999 |
| TOX21 | 793 | 5399 | 0.868 | 0.881 | 0.339 | 0.347 |

Table 1: Class breakdown and evaluation metrics for each dataset. $N_p$ and $N_n$ denote the number of positive and negative examples respectively.

used pretrained Inception V3 network and extract deep features of the underlying image for bounding boxes for all the nodes of a scene-graph



# Explanations on Molecular Graphs

Dataset : BBBP, BACE, NR-ER from TOX21

- BBBP : whether a permeates the human blood brain barrier
- BACE : whether a molecule inhibits the human enzyme b-secretase
- TOX21 : measurements of molecules for several toxicity targets

| Dataset | $N_p$ | $N_n$ | AUC-ROC | | AUC-PR | |
|---|---|---|---|---|---|---|
| | | | Train | Test | Train | Test |
| Country vs. Urban | 3267 | 3862 | 0.987 | 0.986 | 0.939 | 0.936 |
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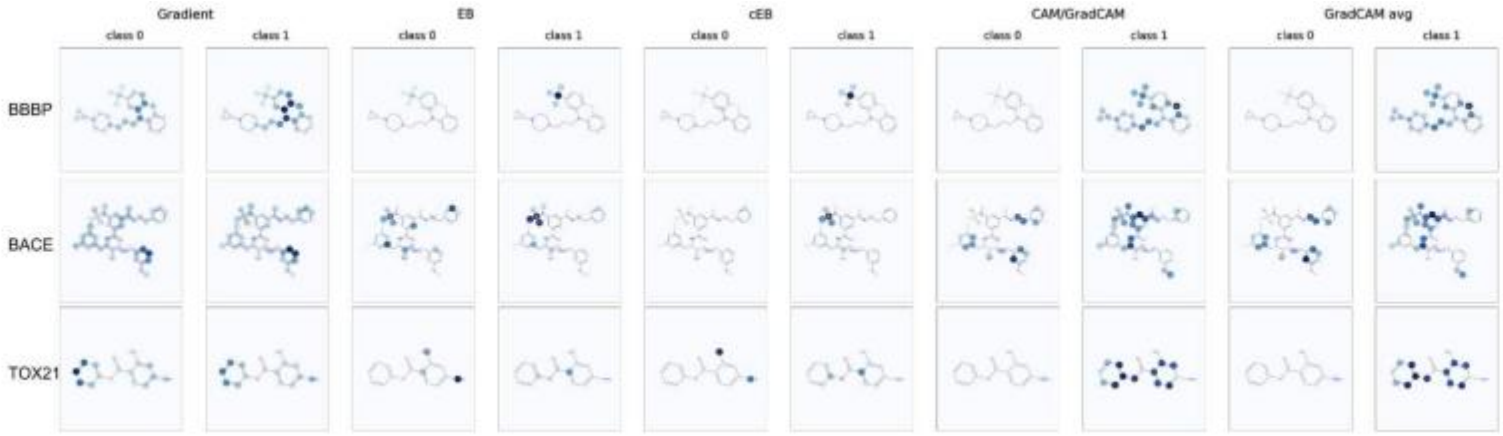
Table 1: Class breakdown and evaluation metrics for each dataset. $N_p$ and $N_n$ denote the number of positive and negative examples respectively.

Selected explanation results for each molecule dataset

A darker blue color indicates a higher relevance for a given class



# Analysis of Explanation Methods

**Fidelity** : calculated to capture the intuition that occlusion of salient features identified through explanations should decrease classification accuracy

**Contrastivity** : designed to capture the intuition that class-specific features highlighted by an explanation method should differ between classes

**Sparsity** : designed to measure the localization of an explanation

| | Country vs. Urban | Indoor vs. Outdoor | BBBP | BACE | TOX21 | |
|---|---|---|---|---|---|---|
| **CG** | 0.40 | 0.40 | 0.19 | 0.38 | 0.53 | **Fidelity** |
| | 0.02 ± 0.45 | 0.18 ± 1.88 | 0.45± 2.19 | 0.77± 2.99 | 0.2 ± 2.13 | **Contrastivity** |
| | 11.67 ± 16.42 | 14.23 ± 17.45 | 0.22± 2.43 | 0.28±1.58 | 0.21±2.98 | **Sparsity** |
| **CAM/ Grad-CAM** | 0.25 | 0.25 | 0.17 | 0.36 | 0.11 | **Fidelity** |
| | 99.99 ± 0.39 | 100.00 ± 0.11 | 99.99± 0.11 | 100.0± 0.0 | 99.99± 0.29 | **Contrastivity** |
| | 2.82 ± 7.84 | 2.62 ± 8.07 | 6.26±7.83 | 9.36±7.67 | 4.86±8.74 | **Sparsity** |
| **Grad-CAM Avg** | 0.26 | 0.24 | 0.17 | 0.38 | 0.17 | **Fidelity** |
| | 58.41 ± 19.70 | 59.93 ± 20.47 | 41.06±19.05 | 29.22±14.04 | 44.03±23.7 | **Contrastivity** |
| | 0.57 ± 7.07 | 0.63 ± 7.30 | 0.01±0.07 | 0.0±0.0 | 0.01±0.11 | **Sparsity** |
| **EB** | 0.12 | 0.06 | 0.18 | 0.38 | 0.19 | **Fidelity** |
| | 29.68 ± 25.45 | 68.35 ± 36.35 | 50.87±18.76 | 60.29±15.40 | 49.06±22.59 | **Contrastivity** |
| | 60.92 ± 23.28 | 64.31 ± 22.48 | 40.35±22.11 | 51.4±13.97 | 30.12±23.04 | **Sparsity** |
| **c-EB** | 0.07 | 0.04 | 0.18 | 0.35 | 0.12 | **Fidelity** |
| | 81.46 ± 21.69 | 73.41 ± 37.76 | 96.97± 5.68 | 97.04±5.12 | 97.23±9.3 | **Contrastivity** |
| | 59.35 ± 23.13 | 76.31 ± 25.15 | 40.54±21.69 | 53.01±13.95 | 31.31±22.91 | **Sparsity** |

- extend explainability methods designed for CNNs to GCNNs and compared these methods **qualitatively** and **quantitatively**
- demonstrated the extended methods on two application domains: **visual scene graph, molecular graph**
- characterized the explanations obtained by each method with three metrics: **fidelity, contrastivity, sparsity**

- Grad-CAM is most suitable among the studied methods for explanations on graphs of moderate size
  However, readers need to choose a method with characteristics most appropriate to their explication

Thank you ツ